

Central Forces, Effective Potential, Conic Orbits, and the Runge–Lenz Vector A

Notes prepared by Goutam Sheet for PHY101, 2025

1. Central force basics

A *central force* always points along the radius vector from a fixed center and depends only on the distance r :

$$\mathbf{F}(\mathbf{r}) = F(r) \hat{\mathbf{r}}, \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{r}.$$

Important examples: inverse-square attractions such as gravity or electrostatics for unlike charges:

$$\mathbf{F}(r) = -\frac{k}{r^2} \hat{\mathbf{r}}, \quad k > 0.$$

Because $\mathbf{F}$ is strictly radial, it exerts no torque about the origin:

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{0}.$$

Hence the angular momentum

$$\mathbf{L} = m \mathbf{r} \times \mathbf{v}$$

is conserved. The motion is planar (in a plane perpendicular to $\mathbf{L}$), and the areal velocity is constant: $\frac{1}{2}r^2\dot{\theta} = \text{const}$ (Kepler's second law).

Working in polar coordinates (r, θ) in the plane,

$$v^2 = \dot{r}^2 + (r\dot{\theta})^2, \quad L = mr^2\dot{\theta} \Rightarrow \dot{\theta} = \frac{L}{mr^2}.$$

2. Energy and the effective potential

For the attractive inverse-square force with potential $V(r) = -k/r$ (zero at infinity), the total mechanical energy is

$$E = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}m(r\dot{\theta})^2 - \frac{k}{r} = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} - \frac{k}{r}.$$

Define the *effective potential*

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{k}{r},$$

so that the radial motion obeys

$$\frac{1}{2}m\dot{r}^2 + V_{\text{eff}}(r) = E.$$

2.1 Reading $V_{\text{eff}}(r)$ without plotting (I am sure you remember the plot made in the class)

Key features:

- As $r \rightarrow 0$, $V_{\text{eff}} \rightarrow +\infty$ (centrifugal wall).
- As $r \rightarrow \infty$, $V_{\text{eff}} \rightarrow 0^-$.
- The minimum occurs where $\frac{dV_{\text{eff}}}{dr} = 0$:

$$-\frac{L^2}{mr^3} + \frac{k}{r^2} = 0 \Rightarrow r_0 = \frac{L^2}{mk}.$$

This corresponds to a *stable circular orbit* with energy

$$E_{\text{circ}} = V_{\text{eff}}(r_0) = \frac{L^2}{2mr_0^2} - \frac{k}{r_0} = -\frac{mk^2}{2L^2}.$$

Comparing a horizontal energy line E with V_{eff} classifies the orbit:

$$\begin{aligned} E = E_{\text{circ}} &\Rightarrow \text{circle at } r = r_0, \\ E_{\text{circ}} < E < 0 &\Rightarrow \text{ellipse (bound)}, \\ E = 0 &\Rightarrow \text{parabola (threshold)}, \\ E > 0 &\Rightarrow \text{hyperbola (unbound)}. \end{aligned}$$

Turning points (periapsis/apoapsis) satisfy $E = V_{\text{eff}}(r)$:

$$E = \frac{L^2}{2mr^2} - \frac{k}{r} \Rightarrow 2mEr^2 + 2mkr - L^2 = 0,$$

with roots

$$r_{\pm} = \frac{-mk \pm \sqrt{m^2k^2 + 2mEL^2}}{2mE} \quad (E \leq 0 \text{ gives real, positive turning radii}).$$

3. Orbit equation $r(\theta)$: two Newtonian derivations

We will use l for the semi-latus rectum. The target result is

$$\boxed{r(\theta) = \frac{l}{1 + e \cos(\theta - \theta_0)}}, \quad \boxed{l = \frac{L^2}{mk}}.$$

3.1 Method A: Energy + the substitution $u = 1/r$

Start from

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} - \frac{k}{r}, \quad \dot{\theta} = \frac{L}{mr^2}.$$

Eliminate time using

$$\dot{r} = \frac{dr}{dt} = \frac{dr}{d\theta} \frac{d\theta}{dt} = \frac{dr}{d\theta} \frac{L}{mr^2} \Rightarrow \dot{r}^2 = \left(\frac{L}{mr^2} \frac{dr}{d\theta} \right)^2.$$

Substitute and multiply by $\frac{2mr^2}{L^2}$:

$$\left(\frac{dr}{d\theta} \right)^2 \frac{1}{r^2} + 1 - \frac{2kr}{L^2} = \frac{2Er^2}{L^2}.$$

Let $u = 1/r$ so that $dr/d\theta = -(1/u^2) du/d\theta$. Then

$$\left(\frac{dr}{d\theta} \right)^2 \frac{1}{r^2} = \left(\frac{du}{d\theta} \right)^2,$$

and we get

$$\left(\frac{du}{d\theta} \right)^2 + u^2 = \frac{2k}{L^2}u + \frac{2E}{L^2}.$$

Differentiate with respect to θ :

$$2 \frac{du}{d\theta} \frac{d^2u}{d\theta^2} + 2u \frac{du}{d\theta} = \frac{2k}{L^2} \frac{du}{d\theta}.$$

Provided $du/d\theta \neq 0$, divide by $2 du/d\theta$:

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{mk}{L^2}}.$$

This linear ODE has solution

$$u(\theta) = \frac{mk}{L^2} \left(1 + e \cos(\theta - \theta_0) \right),$$

hence

$$\boxed{r(\theta) = \frac{l}{1 + e \cos(\theta - \theta_0)}, \quad l = \frac{L^2}{mk}}.$$

3.2 Method B: Directly from Newton's radial equation

Newton's radial equation in polar coordinates is

$$m(\ddot{r} - r\dot{\theta}^2) = -\frac{k}{r^2}, \quad \dot{\theta} = \frac{L}{mr^2}.$$

Thus $r\dot{\theta}^2 = \frac{L^2}{m^2r^3}$ and

$$\ddot{r} - \frac{L^2}{m^2r^3} = -\frac{k}{mr^2}.$$

Let $u = 1/r$. Then

$$\dot{r} = -\frac{L}{m} \frac{du}{d\theta}, \quad \ddot{r} = -\frac{L^2}{m^2} u^2 \frac{d^2u}{d\theta^2}, \quad r\dot{\theta}^2 = \frac{L^2}{m^2} u^3.$$

Substituting,

$$-\frac{L^2}{m^2} u^2 \frac{d^2u}{d\theta^2} - \frac{L^2}{m^2} u^3 = -\frac{k}{m} u^2 \Rightarrow \boxed{\frac{d^2u}{d\theta^2} + u = \frac{mk}{L^2}}.$$

Solve as above to obtain the same conic section with $l = L^2/(mk)$.

4. The Runge–Lenz vector $\mathbf{A}$: definition, conservation, and orbit

4.1 Definition

$$\boxed{\mathbf{A} = \mathbf{p} \times \mathbf{L} - mk \hat{\mathbf{r}}}, \quad \mathbf{p} = m\mathbf{v}.$$

It is conserved for the inverse-square law and points toward periapsis. Its magnitude gives the eccentricity via $|\mathbf{A}| = mk e$.

4.2 Proof that $\frac{d\mathbf{A}}{dt} = 0$

Differentiate:

$$\frac{d\mathbf{A}}{dt} = \frac{d\mathbf{p}}{dt} \times \mathbf{L} + \mathbf{p} \times \frac{d\mathbf{L}}{dt} - mk \frac{d\hat{\mathbf{r}}}{dt}.$$

(i) Since $\mathbf{L} = m\mathbf{r} \times \mathbf{v}$, and $\mathbf{F}$ is central,

$$\frac{d\mathbf{L}}{dt} = m(\mathbf{r} \times \dot{\mathbf{v}} + \dot{\mathbf{r}} \times \mathbf{v}) = m\left(\mathbf{r} \times \frac{\mathbf{F}}{m} + \mathbf{v} \times \mathbf{v}\right) = \mathbf{0}.$$

(ii) With $\dot{\mathbf{p}} = \mathbf{F} = -\frac{k}{r^2}\hat{\mathbf{r}}$,

$$\frac{d\mathbf{p}}{dt} \times \mathbf{L} = -\frac{k}{r^2} \hat{\mathbf{r}} \times \mathbf{L}.$$

Use $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b})$ with $\mathbf{a} = \hat{\mathbf{r}}$, $\mathbf{b} = \mathbf{r}$, $\mathbf{c} = \mathbf{p}$:

$$\hat{\mathbf{r}} \times \mathbf{L} = \hat{\mathbf{r}} \times (\mathbf{r} \times \mathbf{p}) = \mathbf{r}(\hat{\mathbf{r}} \cdot \mathbf{p}) - \mathbf{p}(\hat{\mathbf{r}} \cdot \mathbf{r}) = m(\mathbf{r} \dot{r} - r \mathbf{v}).$$

Hence

$$\frac{d\mathbf{p}}{dt} \times \mathbf{L} = -\frac{k}{r^2} m(\mathbf{r} \dot{r} - r \mathbf{v}) = -\frac{mk\dot{r}}{r^2} \mathbf{r} + \frac{mk}{r} \mathbf{v}.$$

(iii) From $\mathbf{r} = r\hat{\mathbf{r}}$,

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r} \hat{\mathbf{r}} + r \frac{d\hat{\mathbf{r}}}{dt} \Rightarrow \frac{d\hat{\mathbf{r}}}{dt} = \frac{1}{r} \mathbf{v} - \frac{\dot{r}}{r} \hat{\mathbf{r}} = \frac{1}{r} \mathbf{v} - \frac{\dot{r}}{r^2} \mathbf{r}.$$

(iv) Assemble the pieces:

$$\frac{d\mathbf{A}}{dt} = \left[-\frac{mk\dot{r}}{r^2} \mathbf{r} + \frac{mk}{r} \mathbf{v} \right] - mk \left[\frac{1}{r} \mathbf{v} - \frac{\dot{r}}{r^2} \mathbf{r} \right] = \mathbf{0}.$$

Thus $\boxed{\frac{d\mathbf{A}}{dt} = 0}$.

4.3 Orbit from $\mathbf{A} \cdot \mathbf{r}$

Compute

$$\mathbf{A} \cdot \mathbf{r} = (\mathbf{p} \times \mathbf{L}) \cdot \mathbf{r} - mk \hat{\mathbf{r}} \cdot \mathbf{r}.$$

Using the scalar triple product,

$$(\mathbf{p} \times \mathbf{L}) \cdot \mathbf{r} = \mathbf{L} \cdot (\mathbf{r} \times \mathbf{p}) = \mathbf{L} \cdot \mathbf{L} = L^2, \quad \hat{\mathbf{r}} \cdot \mathbf{r} = r,$$

we get

$$\boxed{\mathbf{A} \cdot \mathbf{r} = L^2 - mk r}.$$

Choose the polar angle θ measured from the direction of $\mathbf{A}$ so that $\mathbf{A} \cdot \mathbf{r} = |\mathbf{A}| r \cos \theta$. Then

$$|\mathbf{A}| r \cos \theta = L^2 - mk r \Rightarrow r = \frac{L^2/(mk)}{1 + \frac{|\mathbf{A}|}{mk} \cos \theta}.$$

Identify

$$\boxed{l = \frac{L^2}{mk}}, \quad \boxed{e = \frac{|\mathbf{A}|}{mk}}, \quad \Rightarrow \quad \boxed{r(\theta) = \frac{l}{1 + e \cos \theta}}.$$

5. Eccentricity–energy relations

5.1 Method I: Conic geometry + l

For a conic, $l = a(1 - e^2)$ (ellipse case), so $a = \frac{l}{1 - e^2} = \frac{L^2}{mk(1 - e^2)}$. The orbital energy is

$$E = -\frac{k}{2a} = -\frac{k}{2} \cdot \frac{mk(1 - e^2)}{L^2} = -\frac{mk^2}{2L^2}(1 - e^2).$$

Hence

$$\boxed{e^2 = 1 + \frac{2EL^2}{mk^2}}.$$

This holds for all conics: $E < 0 \Rightarrow e < 1$ (ellipse), $E = 0 \Rightarrow e = 1$ (parabola), $E > 0 \Rightarrow e > 1$ (hyperbola).

5.2 Method II: Magnitude of $\mathbf{A}$

From $\mathbf{A} = \mathbf{p} \times \mathbf{L} - mk \hat{\mathbf{r}}$, one finds the standard identity

$$\boxed{|\mathbf{A}|^2 = m^2 k^2 + 2EL^2}.$$

Therefore

$$\boxed{e = \frac{|\mathbf{A}|}{mk} = \sqrt{1 + \frac{2EL^2}{mk^2}}}.$$

(An efficient verification is obtained by evaluating at periapsis or apoapsis, where $\mathbf{v} \perp \hat{\mathbf{r}}$.)

6. Orbital types via the effective potential

- **Circular orbit:** $r_0 = \frac{L^2}{mk}$, $E_{\text{circ}} = -\frac{mk^2}{2L^2}$, $e = 0$.
- **Ellipses:** $E_{\text{circ}} < E < 0$ so $0 < e < 1$, with turning radii $r_{\min} = \frac{l}{1+e}$ and $r_{\max} = \frac{l}{1-e}$.
- **Parabola:** $E = 0$ so $e = 1$, marginal escape/capture, $r \rightarrow \infty$ at zero asymptotic speed.
- **Hyperbola:** $E > 0$ so $e > 1$, unbound flyby with nonzero asymptotic speed.

Since $e = \sqrt{1 + \frac{2EL^2}{mk^2}}$, increasing E at fixed L increases e : circle $\rightarrow$ ellipse $\rightarrow$ parabola $\rightarrow$ hyperbola.

7. Important results

$$\mathbf{F} = -\frac{k}{r^2}\hat{\mathbf{r}}, \quad V(r) = -\frac{k}{r}$$

$$\mathbf{L} = m\mathbf{r} \times \mathbf{v} = \text{const}, \quad \dot{\theta} = \frac{L}{mr^2}$$

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} - \frac{k}{r}, \quad r_0 = \frac{L^2}{mk}, \quad E_{\text{circ}} = -\frac{mk^2}{2L^2}$$

$$r(\theta) = \frac{l}{1 + e \cos(\theta - \theta_0)}, \quad l = \frac{L^2}{mk}$$

$$\mathbf{A} = \mathbf{p} \times \mathbf{L} - mk\hat{\mathbf{r}}, \quad \frac{d\mathbf{A}}{dt} = 0, \quad e = \frac{|\mathbf{A}|}{mk}$$

$$e = \sqrt{1 + \frac{2EL^2}{mk^2}}$$

Discussions: For in-person questions or to schedule a meeting, email goutam@iisermohali.ac.in. Please indicate whether you prefer a one-on-one conversation or a group discussion.